L2.5 Curve Sketching

§3.5 — precalc + the derivatives table

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1. Sketch the graph of $y = (x-2)^2(x+4)$.

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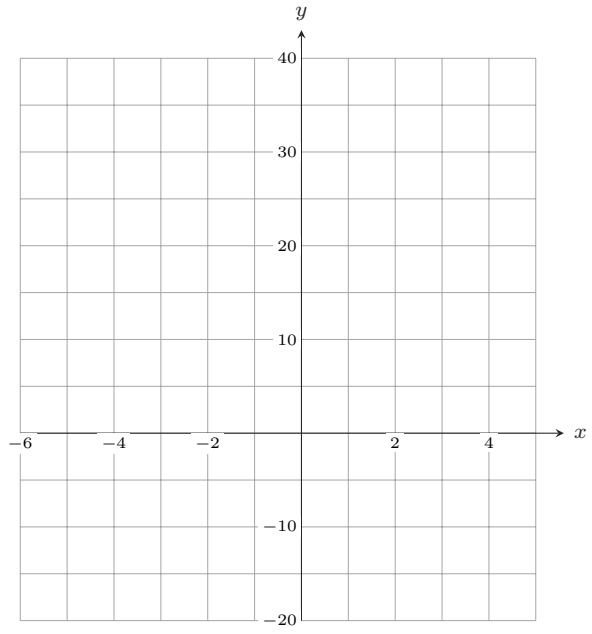
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Derivatives table

$f(x)$	
$f'(x)$	
	⟷
$f''(x)$	



2. Sketch the graph of $f(x) = \sqrt[3]{x}\,(x-4)$.

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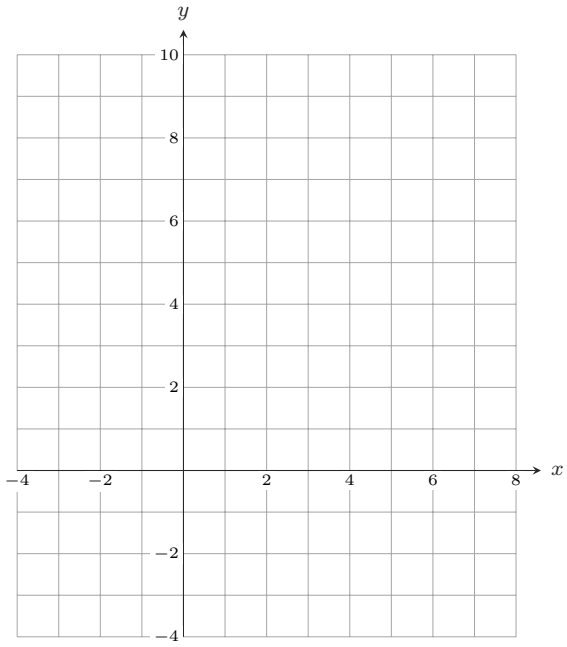
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Derivatives table

$f(x)$	
$f'(x)$	
	$\longleftrightarrow$
$f''(x)$	



3. Sketch the graph of $y = 1 + \dfrac{2}{x} + \dfrac{1}{x^2}$.

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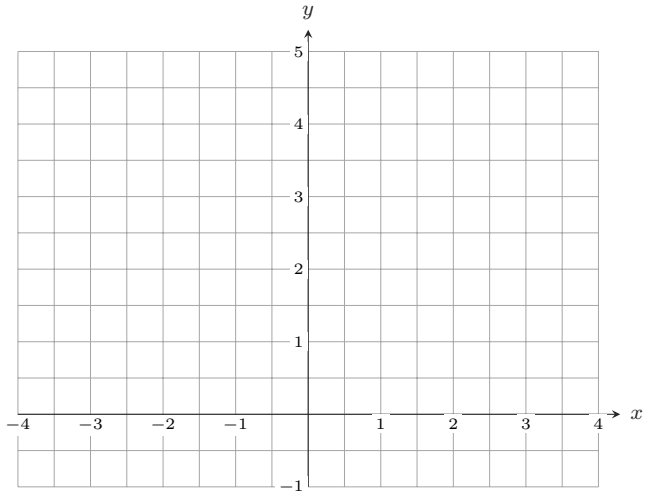
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Derivatives table

$f(x)$	
$f'(x)$	
$f''(x)$	



4. Sketch the graph of $y = \dfrac{x^2 - 2x + 4}{x - 2}$.

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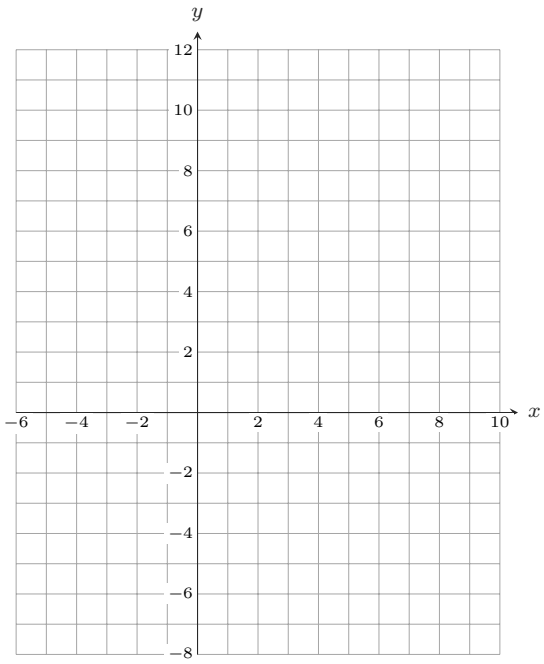
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Derivatives table

$f(x)$	
$f'(x)$	
	$\longleftrightarrow$
$f''(x)$	



5. BOARDS **One rational per group. Sketch it completely on your wall.**

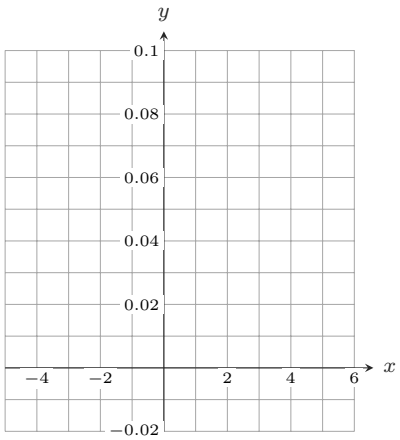
Group 1 $y = \dfrac{1}{(x-3)^2(x+2)^2}$

Group 2 $y = \dfrac{3x^2 - 8x - 3}{x^3 - x}$

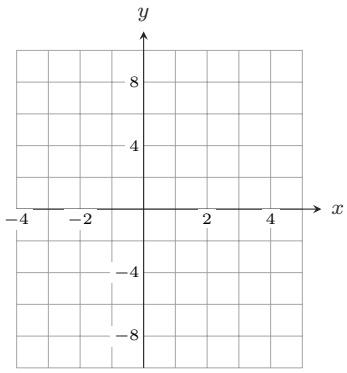
Group 3 $y = \dfrac{(x+2)^2(x-5)}{(x-3)(x+1)(x+4)}$

Group 4 $y = \dfrac{2x^3 - 5x^2 - 11x - 4}{3x^3 + 11x^2 + 5x - 3}$

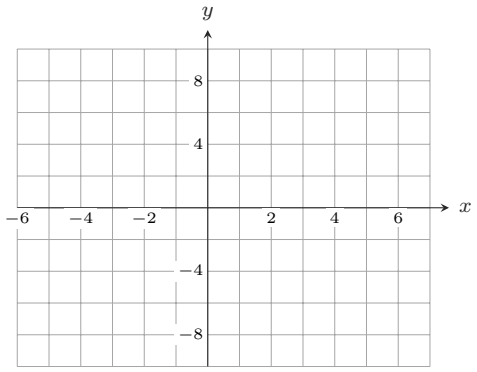
Group 1



Group 2



Group 3



Group 4

